Machine Learning
Quiz Refresher: Chapter 4, Questions

Discriminative and generative classification: sigmoid and softmax, cross-entropy, Gaussian class models, LDA and QDA, naive Bayes, imbalance and prior shift; answers in the separate answer sheet

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2026-10-28

How to use this document

Three rounds, answers in the separate answer sheet. Round A is 20 one-line recall questions. Round B is eight multiple-choice questions, letter plus a one-line justification. Round C is five exam-style problems: a logistic classifier by hand, a one-dimensional LDA/QDA fit, a naive Bayes spam filter, a prior-shift correction and a model-choice discussion. Every number is new. Budget 20 minutes for A and B, 60 minutes for C. Keep $\sigma(s) = 1/(1 + e^{-s})$ and ln values to three decimals.

Round A: rapid recall

A1. Name the two routes to the posterior $p(y \mid x)$ and say what each one models.

A2. Write the sigmoid. State $\sigma(-s)$ in terms of $\sigma(s)$ and say where the decision boundary is at equal costs.

A3. With a cost threshold t on the probability, where does the boundary move in score space?

A4. Write the softmax. Why does adding the same constant to all scores change nothing? What does softmax become for $C = 2$?

A5. Write the one-hot cross-entropy loss of one example and the negative conditional log-likelihood of the whole sample. How are they related?

A6. If the scores are affine in x, what shape are the decision boundaries? Can unequal costs bend them?

A7. Name three things the discriminative objective does not provide.

A8. Write the generative story for one labelled example and the joint distribution it defines.

A9. Write Bayes' rule for $p_\phi(y = c \mid x)$ and express it as a softmax. What is the generative "score"?

A10. Why does the generative log-likelihood split into one term per class? What is the MLE of π_c?

A11. Write the MLE of the class mean and class covariance of a Gaussian class model. How many parameters does one $d \times d$ covariance cost?

A12. LDA versus QDA: which assumption differs, which term cancels in LDA, and what shape are the boundaries in each?

A13. State the naive Bayes assumption and the form the score takes because of it.

A14. Write the smoothed multinomial naive Bayes estimate $\hat{\theta}_{cj}$. Why is $\alpha > 0$ necessary?

A15. What is variance smoothing in Gaussian naive Bayes, and what failure does it prevent?

A16. When does a generative classifier beat a discriminative one, and when the reverse? What is "persistent" about misspecification?

A17. Why is class imbalance an estimation issue rather than a decision issue?

A18. Prior shift: give the generative fix and the discriminative fix. What does the second one require of the training posterior?

A19. "Confidence is not familiarity." Explain in one sentence and name the remedy.

A20. How do large language models combine the generative and discriminative routes?

Round B: multiple choice, letter plus justification

Each question is 2 points: 1 for the letter, 1 for a one-line justification.

B1. **Sigmoid.** The score is $s = 2.2$. The probability of class 1 is about:
(a) 0.69 (b) 0.82 (c) 0.90 (d) 0.96

B2. **Moving the boundary.** Misses are four times as costly as false alarms, so the threshold on p is $t = 0.2$. The boundary in score space is at $s =$
(a) -1.39 (b) -0.2 (c) 0 (d) $+1.39$

B3. **Softmax.** Scores $(1, 2, 3)$. The probability of class 3 is about:
(a) 0.50 (b) 0.665 (c) 0.80 (d) 0.333

B4. **Counting parameters.** QDA with $C = 3$ classes and $d = 10$ features. Parameters spent on covariances alone:
(a) 300 (b) 55 (c) 330 (d) 165

B5. **Smoothing.** Multinomial naive Bayes, $V = 1{,}000$ words, class *ham* has 9,000 word occurrences in total and the word *lottery* never appears in it. With $\alpha = 1$, $\hat{\theta}_{\text{ham, lottery}}$ is:
(a) 10^{-4} (b) 0 (c) 1/9,000 (d) 1/1,000

B6. **Prior shift.** Trained on balanced data ($\pi_t = 0.5$), the calibrated posterior at some x is $r_t = 0.5$. At deployment the positive class has prior $\pi_d = 0.1$. The corrected posterior is:
(a) 0.5 (b) 0.9 (c) 0.1 (d) 0.05

B7. **Choosing a route.** A good reason to prefer a generative classifier:
(a) millions of examples and a complicated boundary (b) it is always better calibrated (c) it is immune to model misspecification (d) few examples and a credible model of $p(x \mid y)$

B8. **A confident toaster.** A softmax image classifier trained on cats and dogs outputs 0.99 for "cat" on a photo of a toaster. Best explanation:
(a) the photo is 99% a cat (b) softmax confidence is relative among the known classes and says nothing about whether x is familiar (c) the model overfitted (d) the class priors shifted

Round C: exam-style problems

C1. A logistic classifier by hand [8 pts]

A binary discriminative classifier uses the affine score $s(x) = w^\top x + b$ with $w = (1.5, -2)$, $b = 0.5$.

(a) For $x^{(1)} = (2, 1)$ and $x^{(2)} = (0, 1)$ compute the score and $\hat{p}(y = 1 \mid x)$.

(b) Both examples have label $y = 1$. Compute the cross-entropy loss of each and the empirical risk $\widehat{R}_{\text{CE}}$ on these two. Which example drives the gradient, and why?

(c) Write the equation of the decision boundary at equal costs and classify both points.

(d) Misses now cost 4 and false alarms 1. New threshold on p, new boundary in score space, and the new decisions. Did the boundary change shape?

(e) Show that $\widehat{R}_{\mathrm{CE}}$ equals $-\frac{1}{N}\log L_{\mathrm{cond}}$: what is being maximised?

C2. One-dimensional Gaussian classes: LDA versus QDA [12 pts]

Training data on one feature: class 0: $\{1, 2, 3\}$; class 1: $\{5, 7\}$.

(a) Maximum-likelihood priors, class means and class variances $\hat{\sigma}_c^2$ (MLE denominators).

(b) Pooled variance for LDA: $\hat{\sigma}^2 = \sum_c N_c \hat{\sigma}_c^2 / N$.

(c) QDA scores $s_c(x) = -\frac{(x-\hat{\mu}_c)^2}{2\hat{\sigma}_c^2} - \frac{1}{2}\ln\hat{\sigma}_c^2 + \ln\hat{\pi}_c$ at $x = 4$. Decision, and the posterior of class 1.

(d) LDA scores $s_c(x) = x\hat{\mu}_c/\hat{\sigma}^2 - \hat{\mu}_c^2/(2\hat{\sigma}^2) + \ln\hat{\pi}_c$ at $x = 4$. Decision. Then solve $s_1(x) = s_0(x)$ for the LDA boundary.

(e) The two models disagree at $x = 4$. Explain, from the fitted variances, why QDA leans toward class 1 there.

(f) In $d = 10$ dimensions with $C = 2$, count the parameters of LDA and of QDA (priors, means, covariances). Which one would you fit with $N = 60$ examples, and why?

C3. Naive Bayes spam filter [10 pts]

Vocabulary $V = 4$: *free, meeting, win, report.* Word counts per class over the training mails:

	free	meeting	win	report	total
spam	10	1	6	1	18
ham	2	9	0	7	18

There are 20 spam and 60 ham mails.

(a) Priors $\hat{\pi}$ and smoothed word probabilities $\hat{\theta}_{cj}$ with $\alpha = 1$ (as fractions of 22).

(b) Message m_1 = "win free free". Compute both scores $s_c = \ln\hat{\pi}_c + \sum_j x_j \ln\hat{\theta}_{cj}$ and the posterior of spam.

(c) Message m_2 = "meeting report report". Decision?

(d) Redo (b) with $\alpha = 0$. What happens to the ham score, and why is that dangerous?

(e) State the independence assumption as a formula and give one reason it is wrong for text yet the classifier still works.

C4. Imbalance and prior shift [8 pts]

Fraud is 2% of transactions. To train, the team subsampled legitimate transactions until the training set was balanced ($\pi_t = 0.5$). The discriminative model is calibrated on that training distribution and outputs $r_t(x) = 0.8$ for a transaction.

(a) Why is imbalance an estimation issue? What did subsampling buy, and what did it break?

(b) Correct r_t to the deployment posterior r_d using the odds formula.

(c) A miss costs 20, a false alarm 1. Threshold, and the decision for this transaction before and after correction.

(d) A colleague used a generative model (naive Bayes) instead. What is their one-line fix, and what stays unchanged?

(e) The team reports precision on the balanced validation set. Why will deployment precision be much lower?

C5. Choosing a route [8 pts]

For each scenario pick discriminative or generative (or a combination), name a concrete model and give the decisive reason in one or two sentences.

(a) 300 labelled support tickets, bag-of-words features with $V = 5{,}000$, three categories.

(b) Five million labelled photos, 100 classes, boundaries nobody can describe.

(c) A medical triage model that must flag inputs unlike anything in the training data rather than guess.

(d) A model whose class frequencies change every season while the inputs produced by each class do not.

(e) LDA and logistic regression both produce linear boundaries. In what do they differ, and when would you expect each to win?